

3(a)

feature	prokaryotic cell	eukaryotic cell
circular DNA	✓	✓
80S ribosomes	X	✓
a cell diameter of 20 µm	X	✓

one mark per correct row

if no marks gained, check correct column for 1 mark

3

3(b)

allow enzyme for protein

*any **three** from:*

protease (produced by *S. epidermidis*) is, secreted / released ;

A protease is an extracellular enzyme

protease breaks down proteins (needed for survival) specific to *S. aureus* ; AW

explanation ; e.g. protease active site complementary to substrate (of *S. aureus*)

protease complementary to substrate and form enzyme-substrate complexes

protease breaks down, protein in biofilm / (biofilm) polymers ;

idea that without biofilm the *S. aureus* cannot remain attached to skin cells ;

other example of *S. aureus* protein broken down ;

e.g. cell membrane proteins / receptors / binding sites

AVP ; e.g. *S. aureus* cells need substances from biofilm for survival

3

Question	Answer	Marks
3(c)(i)	<i>allow, bacteria / pathogens / microorganisms, for S. aureus</i> secrete / produce / AW, mucus / mucin ; mucus, traps <i>S. aureus</i> cells / acts as a barrier (to reach cells) ; R virus	2
3(c)(ii)	ciliated epithelial cell ; A ciliated epithelium cell I ciliated cell	1
3(d)(i)	any three from: prevent formation of crosslinks / cross bridges, (between peptidoglycan chains) ; <i>if not gained, allow ecf for vancomycin difference</i> stop / prevent, synthesis / repair, of cell wall ; <i>differences</i> penicillin, binds to / is an inhibitor of, enzyme(s) / transpeptidase(s) (that catalyse formation of cross links) ; ora vancomycin, binds to / acts on / AW, peptidoglycan / cross link, components ; ora <i>idea that</i> vancomycin, blocks access to peptidoglycan component / may prevent binding of enzyme that joins peptidoglycan subunits together ; <i>if no marks gained because of incorrect knowledge of penicillin mechanism of action, allow one mark as ecf for an answer incorporating a feature of vancomycin mechanism of action shown on Fig. 3.2</i>	3

Question	Answer	Marks
3(d)(ii)	any three from: 1 prescribing / take, antibiotics only when (absolutely) necessary ; A examples e.g. do not use for viral infections do not use as preventative medicine 2 make sure, correct / effective, antibiotic(s), prescribed / used ; A only use antibiotic for the prescribed condition 3 complete course / follow instructions for use ; A <i>ref. to</i> DOTS 4 use other antibacterials or develop new, drugs / antibiotics ; 5 reduce / control, antibiotics in, agriculture / animals used for food ; 6 <i>ref. to</i> break transmission cycle / described example ; e.g. vaccines good hygiene in hospitals quarantine 7,8 AVP ; ; e.g check / improve / AW, knowledge of, healthcare professionals / public, qualified report patterns of antibiotic resistance / AW <i>ref. to</i> monitor to check if antibiotic is effective ; <i>ref. to</i> WHO Global Plan to End TB vary antibiotic treatment use a number of different antibiotics at the same time limit / prevent, antibiotics being sold <i>context is control by prescription</i>	3